

Ontology-Grounded Knowledge Graph Construction for Alzheimer's Research

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Introduction & Objectives

We present a semantic similarity-based approach to construct a Knowledge Graph (KG) guided by the **Alzheimer's Disease Ontology (ADO)**.

- **Goal:** Align extracted biomedical concepts to ADO using multimodal embeddings.
- **Innovation:** Density-based clustering (UMAP+HDBSCAN) to identify novel concept groupings for ontology extension.
- **Scale:** Processed 100,847 PubMed abstracts, yielding 6,795 validated triples.

The 6-Stage Pipeline

The pipeline comprises six sequential stages:

- Ontology Parsing:** Extracts terms from ADO (→ 4,616 terms).
- Phrase Extraction:** Biomedical NLP via scispaCy.
- Fast Fuzzy Deduplication:** Blocking strategy (90% threshold).
- Ensemble Embeddings:** Fused representations for ADO matching.
- Clustering & Validation:** UMAP+HDBSCAN for latent discovery.
- LLM Triple Extraction:** GPT-4o-mini relationship generation.

Multi-Model Ensemble Embeddings

To mitigate model-specific bias, we utilize a weighted ensemble.

Model	Focus	Weight
Gemma Medical	Medical Semantics	0.40
BGE-large-en	Dense Retrieval	0.25
Qwen2.5-Coder	General Semantics	0.20
BioBERT-NLI	Bio-Entailment	0.15

The fused vector $\mathbf{e}_{\text{fused}}$ uses a weighted average of projected embeddings.

Clustering & Quality Validation

For medium-confidence matches, we apply UMAP and HDBSCAN.

Cluster Coherence Score (CCS):

$$CCS = 0.40S_{\text{int}} + 0.25D_{\text{lex}} + 0.25R_{\text{pos}} + 0.10D_{\text{ner}}$$

This metric ensures that novel clusters are semantically and linguistically consistent before human review.

LLM-Enhanced Triple Extraction

Extracted relations across 41 types using GPT-4o-mini:

- **Causal:** *causes, inhibits, regulates, triggers*
- **Clinical:** *diagnosed_by, risk_factor_for*
- **Biological:** *phosphorylates, binds_to*

Total Validated Triples: 6,795

Results: Synthetic QA Benchmark

Task Type	Base	Base+KG	Gain
MCQ	31.2	53.5	+22.3%
True/False	48.5	61.0	+12.5%
Definitions	12.7	34.0	+21.3%

KG-augmented RAG provides significant factual gains across all biomedical reasoning tasks.

Conclusions

- KG retrieval substantially improves factual QA performance.
- Clustering successfully identifies latent concepts for ADO expansion.
- Future work will integrate structural graph embeddings.